

Incidental histologic evidence of silica exposure in lung resections from Central Portugal (2019–2024)

Authors and Affiliations

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Abstract

Background: Crystalline silica remains a relevant occupational hazard in Central Portugal, where quarrying, ceramics, and construction industries are long established. Histologic evidence of silica exposure may be incidentally detected in lung resection specimens, but such findings have rarely been characterized in contemporary surgical series.

Methods: We retrospectively reviewed all lobectomies for primary lung carcinoma performed between 2019 and 2024 at a tertiary pathology center. The presence of birefringent crystalline particles and/or silicotic nodules in non-tumorous lung parenchyma under polarized light defined histologic evidence of silica exposure. Demographic, clinical, and occupational data were retrieved.

Results: Among 373 lobectomies, 45 (12%) showed histologic evidence of silica exposure. Three patients had well-formed silicotic nodules—consistent with true silicosis—and all had long-term high-exposure occupations (construction or mining). The remaining 42 showed isolated birefringent silica particles, interpreted as incidental or early exposure changes. Most patients were male (71.7%) and ever-smokers (71%), with a median age of 71 years. COPD or asthma occurred in 20%, and two patients had prior tuberculosis. Cases originated from 22 municipalities in Central Portugal, predominantly areas linked to stone and ceramic industries, showing a non-random geographic pattern.

Conclusions: Histologic evidence of silica exposure—ranging from birefringent particles to silicotic nodules—remains frequent in lung resections from Central Portugal. These findings

highlight ongoing occupational exposure and reinforce the value of pathologic recognition of silica as a marker of cumulative respiratory injury in current lung-cancer practice.

Introduction

Silica-related lung disease remains a preventable yet persistent occupational health problem. Inhalation of crystalline silica particles can lead to chronic inflammation and fibrosis, forming characteristic silicotic nodules that may appear even in clinically silent cases (1–3). Microscopically, silica appears as strongly birefringent, needle-like particles often clustered within macrophages or fibrotic foci under polarized light microscopy (4).

Central Portugal has a long-standing tradition of ceramic and quarrying industries, where occupational exposure to silica dust is well recognized but underreported (5,6). Despite ongoing surveillance of clinical silicosis, histologic detection of incidental silica in resected lungs has not been systematically studied.

Crystalline silica is classified by the International Agency for Research on Cancer (IARC) as carcinogenic to humans (Group 1), based on sufficient evidence for lung cancer in occupational settings (7). Silica exposure is also associated with increased susceptibility to tuberculosis and chronic respiratory disease (3,8).

This study aimed to describe the clinicopathologic and geographic features of lung resections showing histologic evidence of silica exposure, including both true silicosis and incidental findings.

Methods

All lobectomies for primary lung carcinoma performed between January 2019 and December 2024 at the Centro Hospitalar e Universitário de Coimbra (CHUC) were retrospectively reviewed. Clinical, demographic, and occupational data were retrieved from medical records and pathology reports.

Histologic evidence of silica exposure was defined as the presence of birefringent crystalline particles and/or silicotic nodules in non-tumorous lung parenchyma under polarized light (9). Cases with well-formed silicotic nodules were classified as silicosis. Smoking status, comorbidities (tuberculosis, COPD, asthma), and occupation were recorded when available.

Municipality of residence was documented for each case. Geographic visualization (Figure 1) was based on the geoPortal of the National Laboratory of Energy and Geology (LNEG; CRM 2023) (10). Data were analyzed descriptively using standard statistical methods. This study followed institutional policies for anonymized retrospective analysis.

Results

Among 373 lobectomies for primary lung carcinoma, 45 (12%) showed histologic evidence of silica exposure. Three patients exhibited silicotic nodules, confirming true silicosis. All three had high-exposure occupations (two in mining, one in construction). The remaining 42 patients had isolated birefringent silica particles without nodular fibrosis, interpreted as exposure-related but non-diagnostic of silicosis.

Patients with silica exposure had a median age of 71 years (range 52–83) and were predominantly male (71.7%). A history of smoking was present in 71%. Adenocarcinoma was the most frequent histologic type (72.7%), followed by squamous cell carcinoma (13.6%), carcinoid tumors (4.5%), and others (9.2%). COPD or asthma was documented in 9 patients (20%), and 2 had a history of tuberculosis.

Occupational data were available for all silica-positive patients. Sixty percent had worked in construction or stone-related industries, including masonry, quarrying, and ceramics. Others had backgrounds in agriculture, metalwork, or transport. A minority (approximately 15%) held service jobs without known occupational exposure. Geographically, patients originated from 22 municipalities in Central Portugal. Clustering occurred in areas traditionally associated with quarrying, ceramics, and the Panasqueira mining district. This distribution appeared non-random and corresponded to known industrial activity areas, suggesting exposure related to regional occupation rather than incidental detection (5,10,11).

Discussion

Histologic evidence of silica exposure was found in approximately one in eight lobectomies for lung carcinoma, a higher frequency than in most European surgical series (12,13). This likely reflects the persistence of stone and ceramic industries in Central Portugal and the associated occupational burden (5,10).

Although crystalline silica is a Group 1 human carcinogen with sufficient evidence for lung cancer in exposed workers (7,14), its contribution to risk is mainly related to cumulative occupational exposure, particularly in squamous carcinomas (12–15). The predominance of adenocarcinoma in this cohort likely reflects national smoking patterns rather than a silica-driven effect.

The coexistence of COPD and asthma in one-fifth of silica-positive patients suggests overlapping chronic inhalational injury, possibly exacerbated by tobacco exposure (3,8). Although only two patients had previous tuberculosis, this finding aligns with the well-documented association between silica exposure or silicosis and mycobacterial infection (3,8,16).

All cases represented surgically resectable disease (clinical stage I–II). Several patients underwent preoperative EBUS with negative nodal cytology, confirming that silicotic or anthracosilicotic lymphadenopathy was not a confounding factor in staging.

Pathology-based recognition of silica exposure can serve as a practical complement to occupational-health surveillance, particularly in regions where dust-generating industries remain active (5,10,17). Recording such findings in pathology reports may help document the true burden of occupational exposure in current practice.

Conclusion

Silica exposure, detected histologically as birefringent particles or silicotic nodules, remains relatively frequent in lung resections from Central Portugal. These findings highlight the persistence of occupational dust exposure and the diagnostic relevance of recognizing such changes in routine surgical pathology.

Figure legend

Figure 1. Geographic distribution of cases with histologic evidence of silica exposure (2019–2024). Map of Central Portugal showing municipalities of residence of 45 patients with silica-positive lung-resection specimens. Darker shading marks municipalities with more than one case. Clustering corresponds to areas historically involved in stone, ceramic, and mining industries. (High-resolution PNG, 300 dpi, submitted separately.)

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